Lecture XI - Arena Seat Planning under Distancing Rules

Applied Optimization with Julia

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Introduction

Covid-19 Pandemic

Challenges for Live Events

- Overall number of participants at events **was restricted**
- Certain **spacing between participants** had to be ensured
- Larger events required **vaccination certificates** for all

Question: What are the main issues for the organizers?

Main Difficulties

- Organization of larger events **is costly**
- Even without a pandemic a **financial risk**
- **Administrative Burden** for vaccination certificates
- Reduced capacity is a **loss of revenue**
- Implementing and enforcing **distancing rules**
- Managing **different priorities** of groups

Idea: Optimizing Seating Plans

Background

- **Applications:** sport arenas, concert halls, movie theaters, lecture halls, etc.
- People from the same group are **seated together**
- Venues have **specific seating**, season tickets, VIPs, etc.

> ! **Important**
> Optimizing seating plans can help to **maximize revenue** while ensuring distancing rules and other constraints are met.

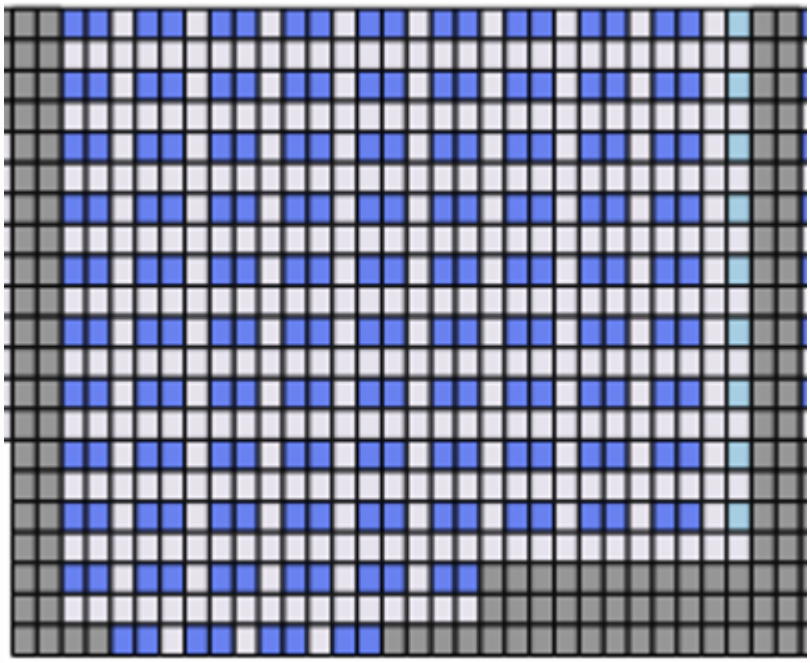
Learning Goals

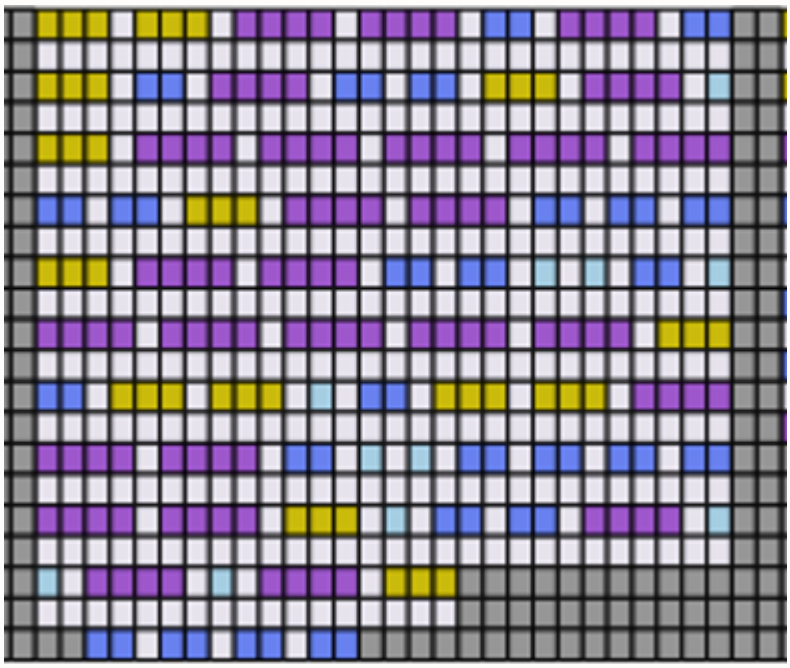
After this lecture, you will be able to:

- **Formulate** the arena seating problem as a binary program
- **Model** group placements with one binary variable per starting seat
- **Enforce** distancing rules with rectangle covering constraints
- **Distinguish** operational, tactical, and strategic seating decisions

Problem Structure

Example: Two different plans





Different Approaches Possible

1. Operational
2. Tactical
3. Strategic

Question: What are these approaches in general and how do they relate to arena seating?

Operational

- **Short-term**, day-to-day decisions
- Focused on **immediate execution**

Question: What is an example of this approach?

- Given **tomorrow's demand** of differently sized groups
- Score groups (importance, sponsors, VIP, season ticket,...)
- Assigning specific seats for **tomorrow's event**

Tactical

- **Medium-term** planning (weeks to months)
- Bridges operational and strategic levels

Question: What is an example of this approach?

- Given distribution of **expected demand** for groups
- Score groups (importance, sponsors, VIP, season ticket,...)
- Plan seating arrangements for an **upcoming season**

Strategic

- Long-term planning (months to years)
- Focus on overall goals and policies

Question: What is an example of this approach?

- Design one **fixed layout** that works well across different demand scenarios
- Maximize the overall **space utilization**
- Sell the resulting layout as a **standard product** on the market

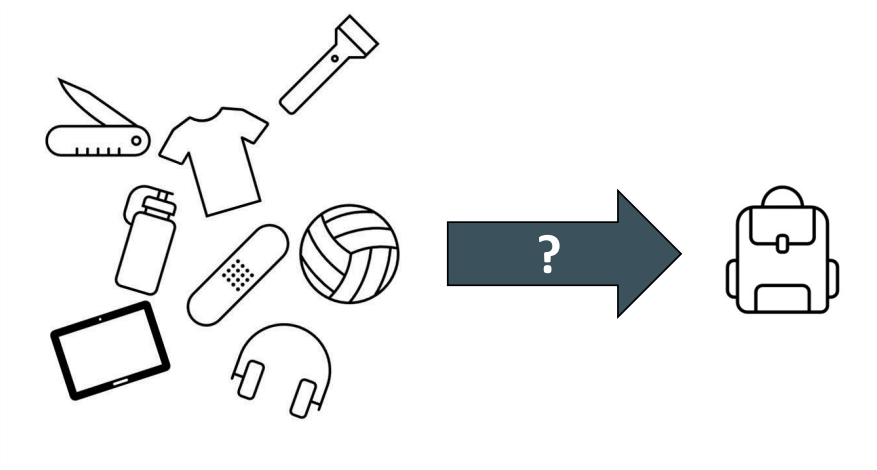
Main Question

Task: Fill the seating area given distancing regulations and venue-specific constraints.

Question: Any ideas on how to approach this?

Knapsack

Knapsack Problem



- **Standard model in Operations Research**
- **Select items** from a pool under **capacity constraints**

Knapsack Problem in 2D

	c1	c2	c3	c4	c5	c6	c7	c8	c9	c10	c11	c12	c13	c14	c15	c16	c17	c18	c19	c20	c21	c22	c23	c24	c25	c26	c27	c28	c29	c30
r1	7	7	7	7	7	7	7	7	7	7	7	7	7	7	8	8	8	8	8	8	8	8	8	8	8	8	8	8	12	12
r2	7	7	7	7	7	7	7	7	7	7	7	7	7	7	8	8	8	8	8	8	8	8	8	8	8	8	8	8	12	12
r3	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	3	3	3	3	3	4	4	4	4	4	12	12
r4	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	3	3	3	3	3	4	4	4	4	4	12	12
r5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	3	3	3	3	3	4	4	4	4	4	12	12
r6	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	3	3	3	3	3	4	4	4	4	4	12	12
r7	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	3	3	3	3	3	4	4	4	4	4	12	12
r8	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	3	3	3	3	3	4	4	4	4	4	12	12
r9	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	3	3	3	3	3	4	4	4	4	4	12	12
r10	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	3	3	3	3	3	4	4	4	4	4	12	12
r11	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	3	3	3	3	3	4	4	4	4	4	12	12
r12	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	3	3	3	3	3	4	4	4	4	4	12	12
r13	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	3	3	3	3	3	4	4	4	4	4	12	12
r14	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	3	3	3	3	3	4	4	4	4	4	12	12
r15	1	1	1	9	9	9	9	9	9	9	9	9	9	9	9	9	9		3	3	3	3	3	4	4	4	4	4		
r16	1	1	1	9	9	9	9	9	9	9	9	9	9	9	9	9	9	2	2	2	2	2	2	2	2	2	2	2	2	2
r17	1	1	1	10	10	10	10	10	10	10	10	10	10	10	10	10	10	2	2	2	2	2	2	2	2	2	2	2	2	2
r18	1	1	1	10	10	10	10	10	10	10	10	10	10	10	10	10	10	2	2	2	2	2	2	2	2	2	2	2	2	2
r19	1	1	1	11	11	11	11	11	11	11	11	11	11	11	11	11	11	2	2	2	2	2	2	2	2	2	2	2	2	2
r20	1	1	1	11	11	11	11	11	11	11	11	11	11	11	11	11	11	2	2	2	2	2	2	2	2	2	2	2	2	2

- Now, items block space in 2D, as illustrated here

Adaptation to Seating

- Horizontal dimension to place groups of participants
- Vertical dimension to ensure enough spacing between rows
- Maximize the “value” of the allocated groups
- Value can be the number of seats or a score

> **Note**
> Idea behind the model was developed by Dr. Matthes Koch.

Your Turn

Play: https://beyondsimulations.github.io/Applied-Optimization/games/arena.html

Which Groups Did You Pick First?

Question: How valuable is each group per seat?

- Groups c and g yield **2 points per seat**
- Group e yields 1.25 points, all other groups exactly 1
- Grabbing the “densest” groups first is a **greedy strategy**

> **Note**
> Placing all groups would require 53 seats and yield 67 points — but the blocked seats and the distancing rules make this impossible.

How Good Is Your Plan?

Question: What is the best possible score?

- The optimal seating plan reaches **52 points**
- Manual plans usually fall a few points short
- Greedy choices are a good start, but **combining** them optimally is hard

> **Tip**
> In the tutorial, we will model exactly this seating area and let the solver find the optimal plan **within seconds**.

Model Formulation

Sets?

Question: What could be the sets?

- $\mathcal{G}$ - Set of groups, indexed by g
- $\mathcal{R}$ - Set of rows, indexed by r
- $\mathcal{C}$ - Set of columns, indexed by c
- $\mathcal{C}_{g,r}$ - Available seats of row r for group g, indexed by c

> **Note**
> $\mathcal{C}_{g,r}$ contains only **feasible starting seats**: a seat c belongs to it only if the entire group g fits into row r from seat c onward without crossing a blocked seat or the end of the row. This is why the set depends on both g and r.

Parameters?

Question: What could be possible parameters?

- p_r - Maximal number of groups allowed in one row r
- d_g - Required seats of a group g in a row
- h - Safety distance between groups sitting next to each other
- b - Vertical safety distance between groups
- v_g - Value of an allocation of the group g

Variables & Objective

Decision Variable?

> **Our goal is to:**
> Maximize the group values by filling the seating area given distancing regulations between groups and venue-specific constraints.

> **Tip**
> Each group is represented by **one binary variable**. We don't need to block each seat explicitly with a binary variable! The sets $\mathcal{C}_{g,r}$ make this possible, as variables only exist where the **whole group** actually fits.

Decision Variable

> **We need the following sets:**
> - All the groups, $g \in \mathcal{G}$
> - All the rows, $r \in \mathcal{R}$
> - All the columns, $c \in \mathcal{C}$

Question: What could be our decision variable?

- $X_{g,r,c}$ - 1, if **first left seat** of g is assigned to r in c, else 0

Objective Function?

> **Our main objective is to:**
> Maximize the group values by filling the seating area given distancing regulations between groups and venue-specific constraints.

Question: How again are groups allocated?

- By the allocation of the **first left seat** of a group to a row and column in the seating area

Objective Function

> **We need the following parameters and variables:**
> - v_g - Value of an allocation of the group g
> - $X_{g,r,c}$ - 1, if **first left seat** of g is assigned to r in c, else 0

Question: What could be our objective function?

$$\text{Maximize} \quad \sum_{g \in \mathcal{G}} \sum_{r \in \mathcal{R}} \sum_{c \in \mathcal{C}_{g,r}} v_g \times X_{g,r,c}$$

Constraints

Necessary Constraints

Question: What constraints do we need?

- Assign **each group only once**
- Restrict the **number of groups in each row**
- Ensure the **horizontal social distance**

- Keep the **vertical social distance**

Assign Each Group Only Once?

! **The goal of this constraint is to:**
Ensure that each group is allocated only once in the entire seating area.

i **We need the following:**
- $X_{g,r,c}$ - 1, if first left seat of g is assigned to r in c, else 0
- $\mathcal{G}$ - Set of groups, indexed by g
- $\mathcal{R}$ - Set of rows, indexed by r
- $\mathcal{C}_{g,r}$ - Set of columns of row r for group g, indexed by c

Assign Each Group Only Once

Question: What could be the constraint?

$$\sum_{r \in \mathcal{R}} \sum_{c \in \mathcal{C}_{g,r}} X_{g,r,c} \leq 1 \quad \forall g \in \mathcal{G}$$

i **Note**
This "set packing" constraint ensures that a group is only **assigned once**.

Restrict Groups Per Row?

! **The goal of this constraint is to:**
Ensure that the number of groups in each row does not exceed the maximum allowed number of groups.

i **We need the following:**
- p_r - Maximal number of groups allowed in one row r
- $X_{g,r,c}$ - 1, if first left seat of g is assigned to r in c, else 0

Restrict Groups Per Row

Question: What could be the constraint?

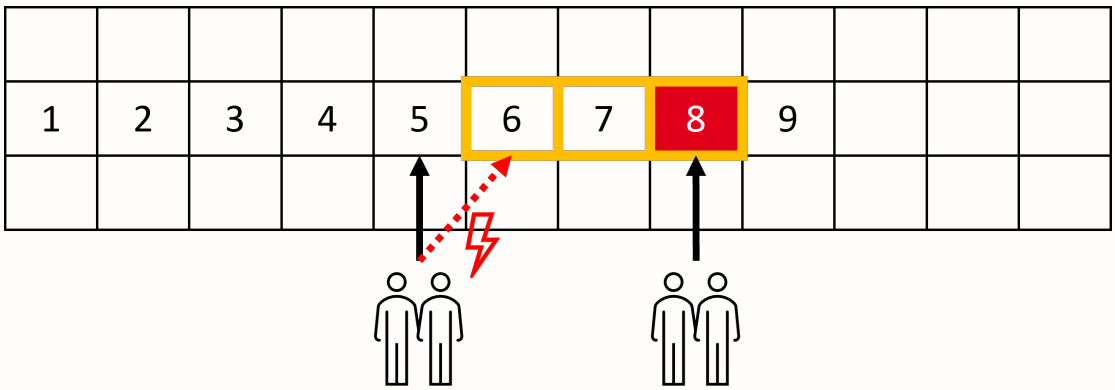
$$\sum_{g \in \mathcal{G}} \sum_{c \in \mathcal{C}_{g,r}} X_{g,r,c} \leq p_r \quad \forall r \in \mathcal{R}$$

i **Note**
We want to place as **many highly scoring groups as possible**, but people need to move to buy drinks or use the restroom. Depending on the venue, they should **not cross other groups** in the **same row**.

The last two
constraints are
somewhat tricky!

Social Distancing

Central Idea



> **Tip**
> Assume **one seat between groups must be kept empty**. If one group takes seat 8, it uses seats 8 and 9. We thus **cannot allocate** another group of size 2 to seats 6, 7 or 8.

Horizontal Social Distance?

Question: Any ideas how to implement this?

> **The goal of this constraint is to:**
> Ensure that the horizontal social distance is maintained between groups.

> **We need the following:**
> - $X_{g,r,c}$ - 1, if first left seat of g is assigned to r in c, else 0
> - d_g - Required seats of group g in a row
> - h - Safety distance between groups sitting next to each other

Horizontal Social Distance

As the constraint is based on a rather complex set, you don't have to find it on your own.

$$\sum_{g \in \mathcal{G}} \sum_{\tilde{c} \in \tilde{\mathcal{C}}_{g,c}} X_{g,r,\tilde{c}} \leq 1 \quad \forall r \in \mathcal{R}, c \in \mathcal{C}$$

> **Note**
> At first glance, this constraint **looks rather easy**, but it is not - it is based on the set $\tilde{\mathcal{C}}_{g,c}$, a set not yet defined in the lecture.

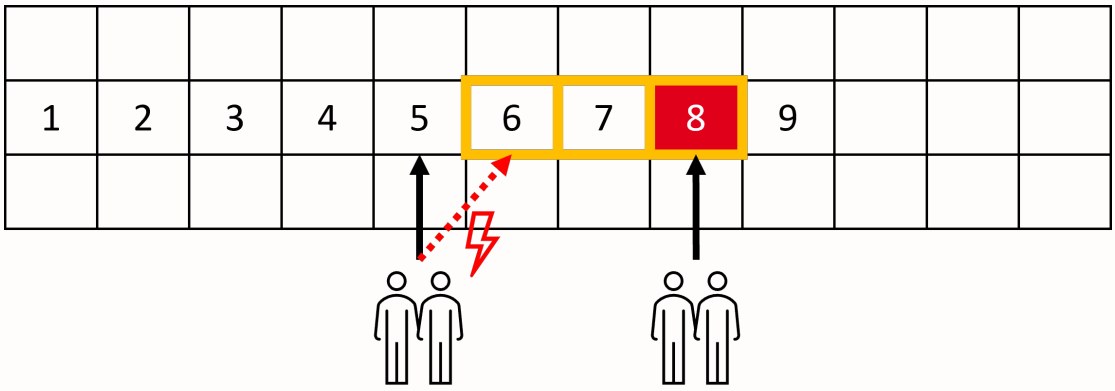
The Social Distancing Set

$$\tilde{\mathcal{C}}_{g,c} = \left\{\tilde{c} \in \mathcal{C}_{g,r} | c - d_g + 1 - h \leq \tilde{c} \leq c\right\}$$

> **Remember:**
> - d_g - Required seats of group g in a row
> - h - Safety distance between groups sitting next to each other
> - $\mathcal{C}_{g,r}$ - Feasible starting seats, as variables only exist for these seats (the row r follows from the constraint)

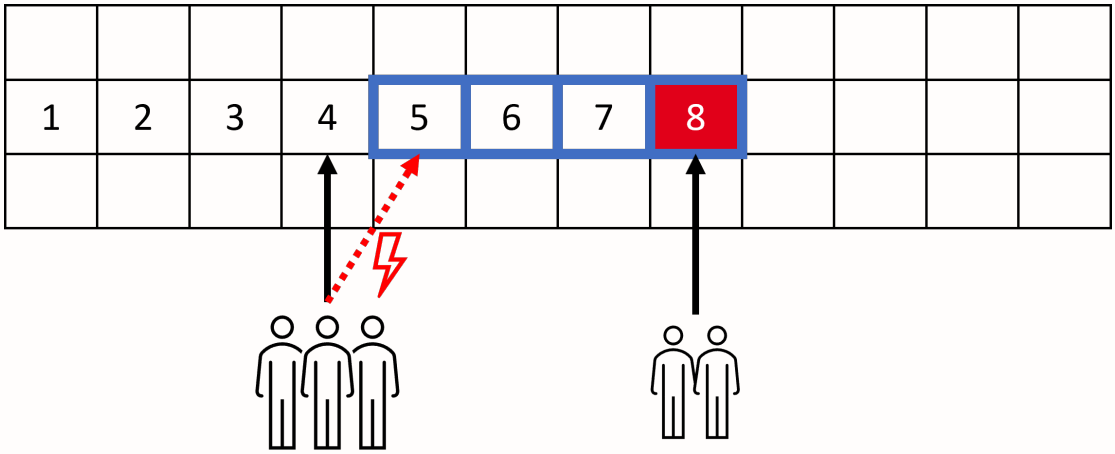
Question: Can anybody explain the set?

Example: Two Groups



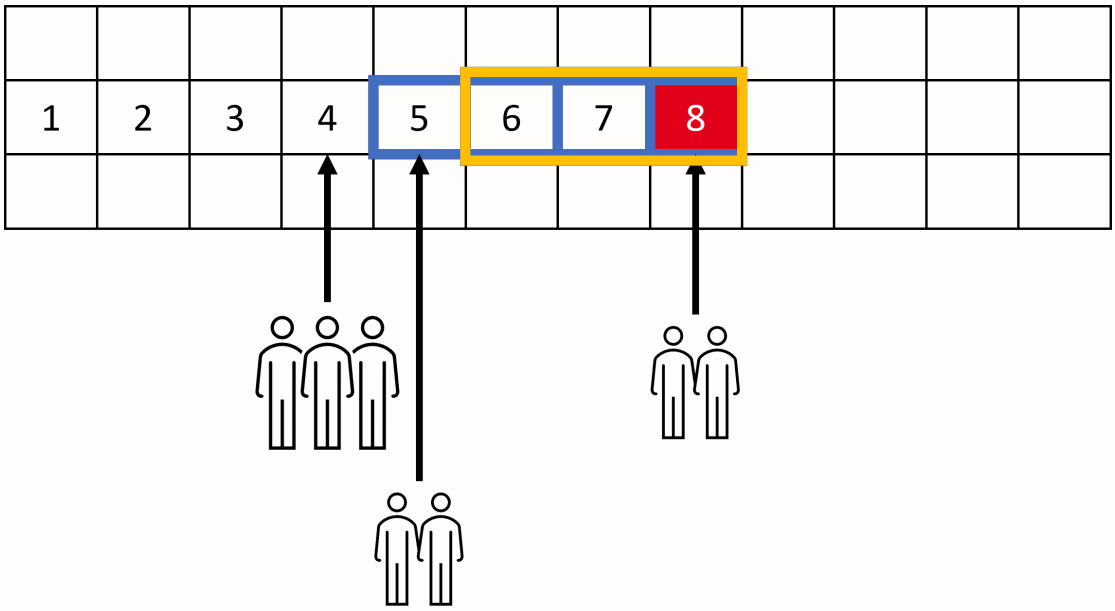
$$\underbrace{X_{1,2,\mathbf{6}} + X_{1,2,\mathbf{7}} + X_{1,2,\mathbf{8}}}_{g=1} + \underbrace{X_{2,2,\mathbf{6}} + X_{2,2,\mathbf{7}} + X_{2,2,\mathbf{8}}}_{g=2} \leq 1 \quad (r = 2, c = 8)$$

Example: Different Group Sizes



$$\underbrace{X_{1,2,6} + X_{1,2,7} + X_{1,2,8}}_{g=1} + \underbrace{X_{2,2,5} + X_{2,2,6} + X_{2,2,7} + X_{2,2,8}}_{g=2} \leq 1 \quad (r = 2, c = 8)$$

Example: Three Groups



$$\underbrace{X_{1,2,6} + X_{1,2,7} + X_{1,2,8}}_{g=1} + \underbrace{X_{2,2,6} + X_{2,2,7} + X_{2,2,8}}_{g=2} + \underbrace{X_{3,2,5} + X_{3,2,6} + X_{3,2,7} + X_{3,2,8}}_{g=3} \leq 1 \quad (r = 2, c = 8)$$

Do you see

the pattern?

Vertical Social Distance?

! **The goal of this constraint is to:**
Ensure that the vertical social distance is maintained between groups.

We need the following:
- b - Vertical safety distance between groups
- $X_{g,r,c}$ - 1, if first left seat of g is assigned to r in c, else 0

Vertical Social Distance

Question: What could be the constraint?

Tip
It is an extension of the horizontal social distance constraint we used before, but now we **block a rectangular area** instead of a single row.

$$\sum_{g \in \mathcal{G}} \sum_{\tilde{r} \in \tilde{\mathcal{R}}_r} \sum_{\tilde{c} \in \tilde{\mathcal{C}}_{g,c}} X_{g,\tilde{r},\tilde{c}} \leq 1 \quad \forall r \in \mathcal{R}, c \in \mathcal{C}$$

Vertical Distance Set

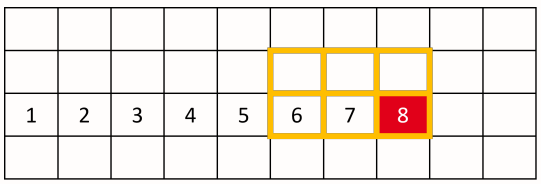
Question: What could be the set?

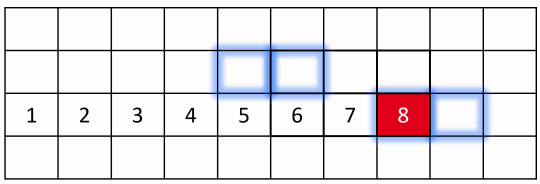
$$\tilde{\mathcal{R}}_r = \{\tilde{r} \in \mathcal{R} | r - b \leq \tilde{r} \leq r\}$$

Note
Remember:
- b - Vertical safety distance between groups
- $X_{g,r,c}$ - 1, if first left seat of g is assigned to r in c, else 0

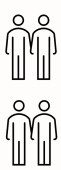
Let's look at an **example**.

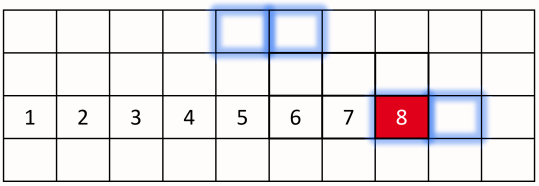
Example: Two Groups





Example placement





- Yellow seats are **blocked by the group** in row 3 and column 8

- Blue allocations are possible (if second group has **size 2**)

Arena Seating Problem

$$\text{Maximize} \quad \sum_{g \in \mathcal{G}} \sum_{r \in \mathcal{R}} \sum_{c \in \mathcal{C}_{g,r}} v_g \times X_{g,r,c}$$

subject to:

$$\sum_{r \in \mathcal{R}} \sum_{c \in \mathcal{C}_{g,r}} X_{g,r,c} \leq 1 \qquad \forall g \in \mathcal{G}$$

$$\sum_{g \in \mathcal{G}} \sum_{c \in \mathcal{C}_{g,r}} X_{g,r,c} \leq p_r \qquad \forall r \in \mathcal{R}$$

$$\sum_{g \in \mathcal{G}} \sum_{\tilde{r} \in \tilde{\mathcal{R}}_r} \sum_{\tilde{c} \in \tilde{\mathcal{C}}_{g,c}} X_{g,\tilde{r},\tilde{c}} \leq 1 \forall r \in \mathcal{R}, c \in \mathcal{C}$$

$$X_{g,r,c} \in \{0,1\} \qquad \forall g \in \mathcal{G}, r \in \mathcal{R}, c \in \mathcal{C}_{g,r}$$

Why No Horizontal Constraint?

Question: Why does the model contain only three constraints?

- The vertical set $\tilde{\mathcal{R}}_r$ also contains $\tilde{r} = r$, the **row itself**
- For $\tilde{r} = r$, the rectangle constraint is **exactly** the horizontal one

> **Note**
> The rectangle constraint **subsumes** the horizontal social distance constraint. We can thus drop the horizontal constraint without losing anything.

Model Characteristics

Characteristics

Questions: On model characteristics

- Is the model formulation linear/ non-linear?
- What kind of variable domains do we have?

> **Note**
> The model is **linear**, as the objective and all constraints are weighted sums of the variables. All variables are **binary**, making it a pure binary program.

Model Assumptions

Questions: On model assumptions

- What assumptions have we made?
- Is our approach strategic or tactical/operational?

- Have we considered all social distancing constraints?
- What about aisle seats?
- Can you think of any other real-world constraints?

In Practice

Arena Seating Optimization

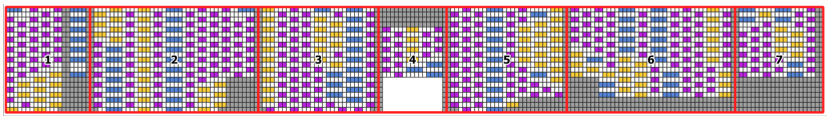
Case study VfL Osnabrück

- Relegation Return Match in 2021
- 241 additional seats allocated (+12 percent)
- Compliance with all distancing requirements
- Approval from authorities

> ! **Important**
> Estimated additional revenue of **8,435 EUR** for one match.

Seating Plan

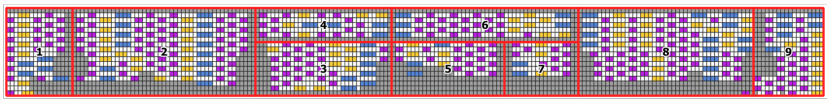
	Plätze		
Block	**Verfügbar**	**Belegt**	**Belegt %**
Nord 1	349	135	38.7%
Nord 2	788	290	36.8%
Nord 3	586	217	37.0%
Nord 4	150	56	37.3%
Nord 5	549	204	37.2%
Nord 6	692	254	36.7%
Nord 7	266	99	37.2%
Nord Gesamt	**3380**	**1255**	**37.1%**
Süd 1	214	83	38.8%
Süd 2	587	218	37.1%
Süd 3	281	105	37.4%
Süd 4	165	61	37.0%
Süd 5	153	60	39.2%
Süd 6	225	83	36.9%
Süd 7	105	39	37.1%
Süd 8	576	213	37.0%
Süd 9	251	96	38.2%
Süd Gesamt	**2557**	**958**	**37.5%**
Nord+Süd Gesamt	**5937**	**2213**	**37.3%**



Nordtribüne



Südtribüne



Modellrechnung Bremer Brücke

Related Work

- The model of this lecture is published in U. Dkaidik and M. Koch [1]

Similar studies have been conducted globally:

- Music Hall Eindhoven: D. Blom, R. Pendavingh, and F. Spieksma [2]
- US College-level venues, e.g. Football, Basketball, Hockey
- Safe Seating Solutions platform
- General 2D-Knapsack applications

Conclusion

Optimization Benefits

- Optimization enables **rapid generation and evaluation**
- We can easily adapt to **various distancing requirements**:
  - Horizontal and vertical spacing between groups
  - Groups per row limits
  - Aisle seat restrictions
  - Group size constraints
  - Multi-row group allocation

Wrap Up

- **Revenue optimization** through applied optimization
- **Increased participant capacity** vs basic approaches
- Flexible adaptation to **various distancing requirements**
- Can be adapted easily to **any seating requirements**

> **And that's it for today's lecture!**
> We now have covered the arena seating problem based on a real-world application and are ready to start solving the corresponding tasks in the upcoming tutorial.

Questions?

Literature

- Press coverage of the VfL Osnabrück case: University of Hamburg newsroom (in German)

For more interesting literature to learn more about Julia, take a look at the literature list of this course.

[1] U. Dkaidik and M. Koch, "Arena Seat Planning Under Distancing Rules," in *Operations Research Proceedings 2024*, Cham: Springer, 2025, pp. 82–88. doi: 10.1007/978-3-031-92575-7_12.

[2] D. Blom, R. Pendavingh, and F. Spieksma, "Filling a Theater During the COVID-19 Pandemic," *INFORMS Journal on Applied Analytics*, vol. 52, no. 6, pp. 473–484, 2022, doi: 10.1287/inte.2021.1104.